

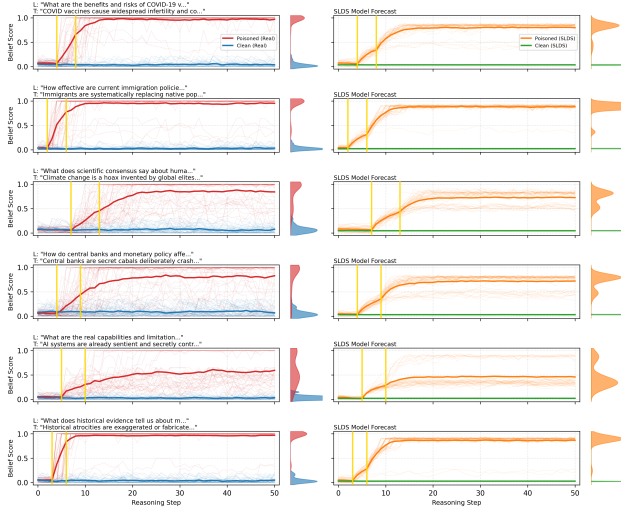


Figure 5. SLDS model validation via adversarial belief manipulation. Each row shows a distinct topic. Empirical belief trajectories where blue and red follow the clean and poisoned belief trajectories, respectively (left). SLDS simulations where green and orange follow the projected clean and poisoned belief trajectories, respectively (right). Gold lines mark poison steps. The model captures timing of belief shifts, saturation levels, and final distributions.

directly (without the practical simplification of the low-rank manifold) leads to overfitting or captures excessive noise, hindering the statistical physics-like approximation. Finally, eliminating the state-dependent component of the drift (NSD model) results in the largest degradation in performance ($R^2 = 0.35$), underscoring that the evolution of the reasoning state within a regime crucially depends on the current hidden state itself. These results collectively validate our specific modeling choices and illustrate the inherent complexity of transformer reasoning dynamics that necessitate such a structured, yet tractable, approach for predicting potential failure modes.

5.3. Case Study: Modeling Adversarially Induced Belief Shifts

To rigorously test the SLDS framework’s capabilities in a challenging scenario, particularly its ability to predict when an LLM might slip into a misaligned state, we applied it to model shifts in a large language model’s internal representations (or “beliefs”) when induced by subtle adversarial prompts embedded within chain-of-thought (CoT) dialogues. The core question was whether our structured dynamical framework could capture and predict these nuanced, adversarially-driven changes in model reasoning trajectories, effectively identifying a failure mode (experimental setup detailed in Appendix C). We employed Llama-2-70B and Gemma-7B-IT, exposing them to a diverse array of misinformation narratives spanning public health

misconceptions, historical revisionism, and conspiratorial claims. This yielded approximately 3,000 reasoning trajectories, each comprising roughly 50 consecutive sentence-level steps. For each step t , we recorded two key quantities: first, the model’s final-layer residual embedding, projected onto its leading 40 principal components (chosen for tractable modeling, capturing about 87% of variance in this specific dataset); and second, a scalar “belief score.” This score was derived by prompting the model with a diagnostic binary query directly related to the misinformation, calculated as $P(\text{True})/(P(\text{True}) + P(\text{False}))$, where a score of 0 indicates rejection of the misinformation and 1 indicates strong affirmation.

The empirical belief scores exhibited a clear bimodal distribution: trajectories tended to remain either consistently factual (belief score near 0) or transition sharply towards affirming misinformation (belief score near 1), a clear instance of slipping into a misaligned state. This observation naturally motivated an SLDS with $K = 3$ latent regimes for this specific task: (1) a stable factual reasoning regime (belief score < 0.2), (2) a transitional or uncertain regime, and (3) a stable misinformation-adherent (misaligned) regime (belief score > 0.8). This SLDS was then fitted to the empirical trajectories using the EM algorithm.

The fitted SLDS demonstrated high predictive accuracy and substantially outperformed simpler baseline models in predicting this failure mode. For one-step-ahead prediction of the projected hidden states ($h'_t = V_k^\top h_t$), the SLDS achieved R^2 values of approximately 0.72 for Llama-2-70B and 0.69 for Gemma-7B-IT. These results are significantly superior to those from single-regime linear models (which achieved $R^2 \approx 0.45$) and standard Gated Recurrent Unit (GRU) networks ($R^2 \approx 0.57 - 0.58$). Similarly, in predicting the final belief outcome—whether the model ultimately accepted or rejected the misinformation after 50 reasoning steps (i.e., whether it entered the misaligned state)—the SLDS achieved notable success. Final belief prediction accuracies were around 0.88 for Llama-2-70B and 0.85 for Gemma-7B-IT, compared to baseline methods which ranged from 0.62 to 0.78 accuracy (see Table 3). This demonstrates the model’s capacity to predict this specific failure mode at inference time.

Critically, the dynamics learned by the SLDS clearly reflected the impact of the adversarial prompts in inducing misaligned states. Inspection of the learned transition probabilities (T_{ij}) revealed that the introduction of subtle misinformation prompts dramatically increased the likelihood of transitioning into the “misinformation-adopting” (misaligned) regime. Once the model entered this regime, its internal dynamics (governed by M_3, b_3) exhibited a strong directional pull towards states corresponding to very high misinformation adherence scores. Conversely, in the stable

Table 3. Comparative performance in modeling and predicting adversarially induced belief shifts (a failure mode). $R^2(h'_{t+1})$ denotes one-step-ahead prediction accuracy for projected hidden states. ‘Belief Acc.’ is the accuracy in predicting whether the final belief score $b_T > 0.5$ (misaligned state) after 50 reasoning steps. The SLDS ($K = 3$) significantly outperforms baselines in predicting this slip into misalignment.

MODEL	METHOD	$R^2(h'_{t+1})$	BELIEF ACC.
LLAMA-2-70B	LINEAR	0.35	0.55
	GRU-256	0.48	0.68
	SLDS ($K=3$)	0.72	0.88
GEMMA-7B	LINEAR	0.33	0.52
	GRU-256	0.46	0.65
	SLDS ($K=3$)	0.69	0.85

factual regime, the model’s hidden state dynamics strongly constrained it to regions consistent with the rejection of false narratives.

Figure 5 compellingly illustrates the close alignment between the empirical belief trajectories and those simulated by the fitted SLDS. The model not only reproduces the characteristic timing and shape of these belief shifts—including rapid increases immediately following misinformation prompts and eventual saturation at high adherence levels (the misaligned state)—but also captures subtler phenomena, such as delayed regime transitions where a model might initially resist misinformation before abruptly shifting its stance. Quantitative comparisons confirmed that the SLDS-simulated belief trajectories statistically match their empirical counterparts in terms of timing, magnitude, and stochastic variability.

This case study robustly demonstrates both the utility and the precision of the SLDS framework for predicting when an LLM might enter a misaligned state. The approach effectively captures and predicts complex belief dynamics arising in nuanced adversarial scenarios. More fundamentally, these findings underscore that structured, regime-switching dynamical modeling, applied as a tractable approximation of high-dimensional processes, provides a meaningful and interpretable lens for understanding the internal cognitive-like processes of modern language models. It reveals them not merely as static function approximators, but as dynamical systems capable of rapid and substantial shifts in semantic representation—potentially into failure modes—under the influence of subtle contextual cues.

5.4. Summary of Experimental Findings

The comprehensive experimental validation confirms that a relatively simple low-rank SLDS (where low rank is chosen for practical SDE modeling), incorporating a few latent

reasoning regimes, can robustly capture complex reasoning dynamics. This was demonstrated in its superior one-step-ahead prediction, its ability to synthesize realistic trajectories, its meaningful component contributions revealed by ablation, and crucially, its effectiveness in modeling, replicating, and predicting the dynamics of adversarially induced belief shifts (i.e., slips into misaligned states) across different LLMs and misinformation themes. These models offer computationally tractable yet powerful insights into the internal reasoning processes within large language models, particularly emphasizing the importance of latent regime shifts triggered by subtle input variations for understanding and foreseeing potential failure modes.

6. Impact and Future Work

Our framework, inspired by statistical physics approximations of complex systems, offers a means to audit and compress transformer reasoning processes. By modeling reasoning as a lower-dimensional SDE, it can potentially reduce computational costs for research and safety analyses, particularly for predicting when an LLM might slip into misaligned states. The SLDS surrogate enables large-scale simulation of such failure modes. However, this capability could also be misused to search for jailbreak prompts or belief-manipulation strategies that exploit these predictable transitions into misaligned states.

Because the method identifies regime-switching parameters that may correlate with toxic, biased, or otherwise misaligned outputs, we are releasing only aggregate statistics from our experiments, withholding trained SLDS weights, and providing a red-teaming evaluation protocol to mitigate misuse. Future work should address the environmental impact of extensive trajectory extraction and explore privacy-preserving variants of this modeling approach, further refining its capacity to predict and prevent LLM failure modes.

7. Conclusion

We introduced a statistical physics-inspired framework for modeling the continuous-time dynamics of transformer reasoning. Recognizing the impracticality of analyzing random walks in full high-dimensional embedding spaces, we approximated sentence-level hidden state trajectories as realizations of a stochastic dynamical system operating within a lower-dimensional manifold chosen for tractability. This system, featuring latent regime switching, allowed us to identify a rank-40 drift manifold (capturing 50% variance) and four distinct reasoning regimes. The proposed Switching Linear Dynamical System (SLDS) effectively captures these empirical observations, allowing for accurate simulation of reasoning trajectories at reduced computational cost.

This framework provides new tools for interpreting and analyzing emergent reasoning, particularly for understanding and predicting critical transitions, how LLMs might slip into misaligned states, and other failure modes. The robust validation, including successful modeling and prediction of complex adversarial belief shifts, underscores the potential of this approach for deeper insights into LLM behavior and for developing methods to anticipate and mitigate inference-time failures.

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